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# Yacht Hydrodynamic Resistance

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1. Data and Problem
2. Log-log Regression
3. BAS / BMA Model Selection
4. Heteroscedasticity-Corrected Model
5. Posterior Predictive
6. Conclusions

Part 1

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# **Data and Problem**

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# The Yacht Hydrodynamics dataset

**Source.** UCI ML Repository — Delft Systematic Yacht Hull Series, towing-tank experiments.

- 22 hull forms, each tested at several speeds
- $n = 308$  observations (one hull  $\times$  one speed per row)
- **Response:** residuary resistance per unit weight of displacement
- **Predictors:** speed (Froude number) + five hull form factors, constant within a hull

## Variables

|              |                        |
|--------------|------------------------|
| LCB          | centre of buoyancy (%) |
| prismatic    | prismatic coefficient  |
| L_disp       | length–displ. ratio    |
| beam_draught | beam–draught ratio     |
| L_beam       | length–beam ratio      |
| Froude       | $Fn = v/\sqrt{gL}$     |
| resistance   | response (target)      |

The five form factors vary only across the 22 hulls;  $Fn \in [0.125, 0.45]$  is the dominant predictor.

Part 2

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## **Log–log Regression**

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# Model and reference prior

**Goal.** Verify the physical power law resistance  $\propto Fn^{\gamma_F}$  with  $\gamma_F \approx 4$ . Taking logs makes the exponent a regression slope:

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(Fn) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

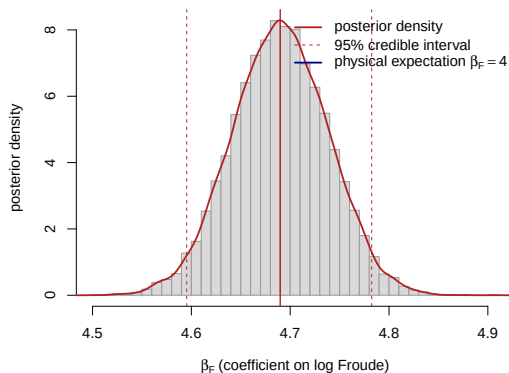
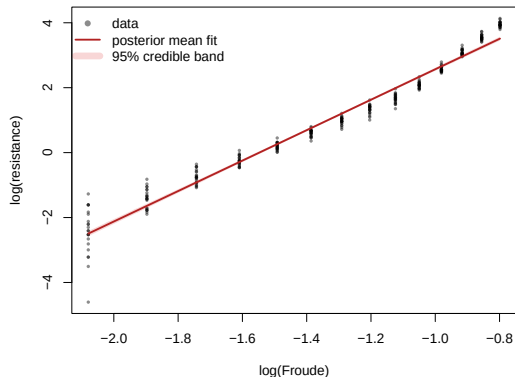
Baseline non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \propto 1, \quad \pi(\sigma^2) \propto \sigma^{-2}.$$

## Closed-form posterior

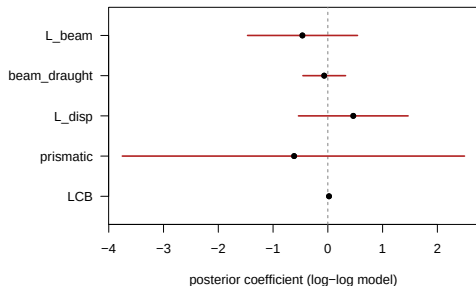
$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$ ; marginalising  $\sigma^2$  gives a multivariate Student- $t$  with  $n - p$  d.o.f. The 95% credible intervals coincide with the OLS confidence intervals — no simulation required.

# Result: the Froude exponent



- Excellent fit:  $R^2 = 0.97$ , residual SD  $s = 0.32$  on the log scale.
- $\beta_F \approx 4.69$ , 95% CI =  $[4.60, 4.78]$ ,  $\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$ .
- Power law confirmed — the exponent is *slightly steeper* than the textbook  $Fn^4$ .

# Hull covariates & residual diagnostics



**Hull form factors.** Once  $Fn$  is in the model, every 95% interval contains zero — the factors vary only across the 22 hulls and are strongly collinear (resolved by BMA in Part 3).

## Residual diagnostics expose two defects:

- U-shaped trend vs. fitted values — a single constant exponent is too rigid;
- spread shrinks as the fitted value grows — textbook *heteroscedasticity*.

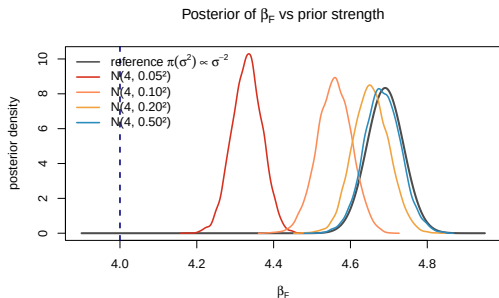
Both motivate the nonlinear correction in Part 4.

# Prior sensitivity: does the prior drive $\beta_F > 4$ ?

Encode the physics in **independent informative priors** and ask how hard they must insist on 4:

$$\beta_F \sim \mathcal{N}(4, \tau^2), \quad \beta_{j \neq F} \sim \mathcal{N}(0, 100^2), \quad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}),$$

sampled with a **Gibbs sampler** ( $10^4$  iters,  $2 \times 10^3$  burn-in), sweeping  $\tau$ .



| Prior on $\beta_F$       | Mean | $\Pr(\beta_F > 4)$ |
|--------------------------|------|--------------------|
| $\mathcal{N}(4, 0.05^2)$ | 4.33 | 1.00               |
| $\mathcal{N}(4, 0.10^2)$ | 4.56 | 1.00               |
| $\mathcal{N}(4, 0.50^2)$ | 4.68 | 1.00               |
| reference prior          | 4.69 | 1.00               |

Only a near-dogmatic  $\tau = 0.05$  drags the mean to 4.33. For any diffuse prior ( $\tau \geq 0.1$ ) the data dominate:  $\beta_F > 4$  is **robust to the prior**.